The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 December 2003.

INCH-POUND

MIL-PRF-19500/406E
8 September 2003
SUPERSEDING
MIL-PRF-19500/406D
29 September 1999

PERFORMANCE SPECIFICATION

* SEMICONDUCTOR DEVICES, DIODE, SILICON, VOLTAGE REGULATOR
TYPES 1N4460, 1N4460C, 1N4460D THROUGH 1N4496, 1N4496C, 1N4496D AND
1N6485, 1N6485C, 1N6485D THROUGH 1N6491, 1N6491C, 1N6491D
1N4460US, 1N4460CUS, 1N4460DUS THROUGH 1N4496US, 1N4496CUS, 1N4496DUS AND
1N6485US, 1N6485CUS, 1N6485DUS THROUGH 1N6491US, 1N6491CUS, 1N6491DUS
PLUS C AND D TOLERANCE SUFFIX; JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for microminiature 1.5 watt silicon, low leakage, voltage regulator diodes with tolerances of 5 percent, 2 percent, and 1 percent. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 (similar to DO-41), 2 (surface mount), 3, 4, and 5 (die).

* 1.3 Maximum ratings. Maximum ratings are as shown in maximum and primary test ratings (see 3.11) herein and as follows: P_T = 1.5 W (derate at 10 mW/°C above T_A = +25°C); -55°C < T_J < +175°C; -65°C < T_{STG} < +175°C

* 1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in maximum and primary test ratings (see 3.11) and as follows:

3.3 V dc ≤ V_Z ≤ 200 V dc (nominal).

a. 1N4460D through 1N4496D and 1N6485D through 1N6491D are 1 percent voltage tolerance.

b. 1N4460C through 1N4496C and 1N6485C through 1N6491C are 2 percent voltage tolerance.

c. 1N4460 through 1N4496 and 1N6485 through 1N6491 are 5 percent voltage tolerance.

$R_{\theta JL}$ = 42°C/W (max) at L = .375 inch (9.52 mm) (nonsurface mount) (see figure 4).
$R_{\theta JEC}$ = 20°C/W (max) (surface mount) (see figure 4).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

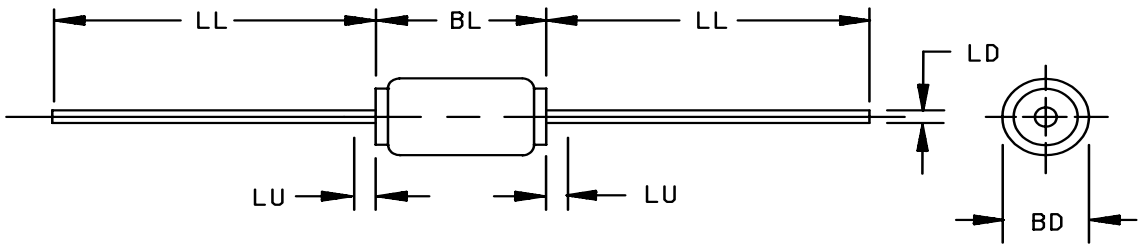
* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500, and as follows:

EC ..... End-cap.
US ..... Surface mount case outline, square end cap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

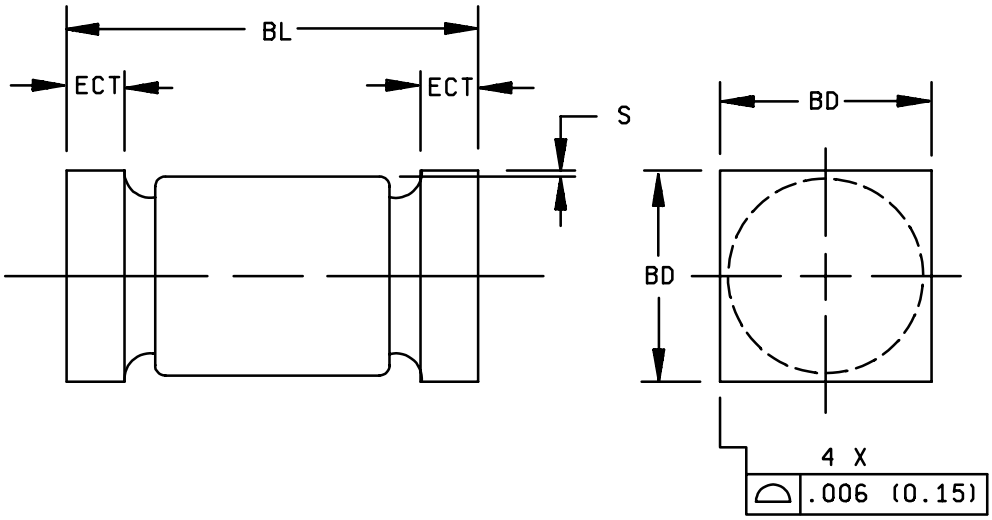


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	.106	.160	2.69	4.06	3
BD	.060	.085	1.52	2.16	3
LL	.800	1.300	20.32	33.02	
LD	.028	.032	0.71	0.81	
LU		.050		1.27	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional with BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. The specified lead diameters apply in the zone between .050 inch (1.27 mm) from the diode body and the end of the lead.
5. Dimensioning and tolerancing shall be in accordance with ASME Y14.5M.

FIGURE 1. Physical dimensions of nonsurface mount device (similar to DO-41).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	.168	.200	4.28	5.08	
ECT	.019	.028	0.48	0.71	
S	.003		0.08		
BD	.091	.103	2.31	2.62	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensioning and tolerancing shall be in accordance with ASME Y14.5M.

FIGURE 2. Physical dimensions of surface mount device, "US".

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2 herein.

3.4.1 Diode construction. These devices shall be constructed utilizing the requirements of MIL-PRF-19500 and as follows.

* 3.4.1.1 Metallurgical bond for diodes with V_Z greater than 6.8 V dc. Category I metallurgical bonds for diodes with V_Z greater than 6.8 V dc as defined in MIL-PRF-19500 shall be utilized.

* 3.4.1.2 Metallurgical bond for diodes with V_Z less than or equal to 6.8 V dc. Category I, or category III metallurgical bonds as defined in MIL-PRF-19500.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

* 3.5.1 Marking of US version devices. For "US" version devices only, all marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container.

* 3.6 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for lead finish the maximum lead temperature shall be 175°C max. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.7 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

3.7.1 Polarity of JANHC and JANKC devices. Polarity marking is not required on JANHC or JANKC devices. All marking shall be retained on the initial container.

* 3.8 Selection of tighter tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, JANTXV, or JANS devices, which have successfully completed all applicable screening, and groups A, B, and C testing as ±5 percent tolerance devices. All sublots of C and D suffix devices shall pass table I, subgroup 2, at tightened tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purposes to establish correlation. For C and D tolerance levels, T_L = 25°C +1°C, -3°C at 0.375 inches (9.53 mm) from body or equivalent.

3.9 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.10 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I and II herein.

* 3.11 Maximum and primary test ratings. Maximum and primary test ratings for voltage regulator diodes are specified in table III herein.

* 3.12 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification Inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Lot accumulation. Lot accumulation period shall be 3 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Supplier imposed requirements shall be documented in the QM plan and must be submitted to the Qualifying Activity for approval. Radiation characterization may be submitted in the QM plan at the option of the manufacturer.

* 4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein shall be performed on the first inspection lot to this revision to maintain qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

* 4.3 Screening (JANS, JANTXV and JANTX levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, appendix E and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurements	
	JANS level	JANTX and JANTXV levels
1a 1b	Required Required	Not required Required (JANTXV only)
2	Not required	Not required
3a 3b (1) 3c	Required Not applicable Thermal impedance, see 4.3.2	Required Not applicable Thermal impedance, see 4.3.2
4	Not applicable	Not applicable
5	Not applicable	Not applicable
6	Not applicable	Not applicable
7	Hermetic seal, gross leak	Hermetic seal, gross leak
8	Required	Not required
9	I_{R1} and V_Z (1N4466 thru 1N4496 only)	Not applicable
10	Required for device > 10 V dc	Not applicable
11	I_{R1} and V_Z $\Delta I_{R1} \leq \pm 100$ percent of initial reading or 50 nA dc, whichever is greater. $\Delta V_Z \leq \pm 2$ percent of initial reading. (2)	I_{R1} and V_Z
12	Required see 4.3.3	Required see 4.3.3
(2) (3) 13	Required Scope display see 4.5.8. Subgroups 2 and 3 of table I herein; ΔI_{R1} (max) $\leq \pm 100$ percent of initial reading or 50 nA, whichever is greater; $\Delta V_Z \leq \pm 2$ percent of initial reading.	Required Subgroup 2 of table I herein; ΔI_{R1} (max) $\leq \pm 100$ percent of initial reading or 50 nA, whichever is greater; $\Delta V_Z \leq \pm 2$ percent of initial reading.
14a 14b	Not applicable Optional	Not applicable Optional
15	Required	Not required
16	Required	Not required

(1) This test shall be performed anytime after screen 3.
(2) Delta limits applicable to 1N4466 thru 1N4496 only.
(3) Thermal impedance not applicable, if already performed 100%.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements) The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum screen limit shall be developed by the supplier using statistical methods and it shall not to exceed the table I, subgroup 2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

* 4.3.3 Power burn-in conditions. Power burn-in conditions are as follows: The test current I_Z shall be adjusted to produce a junction temperature of +125°C minimum and I_Z minimum shall be equal to 50 percent of column 8 of table III. Use method 3100 of MIL-STD-750 to measure T_J (see 4.5.7).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be in accordance with MIL-PRF-19500 and table I herein. Thermal impedance conditions are as follows:

a. I_M measurement current..... 1 mA to 10 mA.

b. I_H forward heating current ..... 3A to 10 A.

c. t_H heating time ..... 10 ms.

d. t_{MD} measurement delay time..... 100 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E table VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein except $Z_{\theta JX}$ need not to be performed.

*4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1056	0°C to +100°C, 10 cycles, n = 22 c = 0.
B3	1051	-55°C to +175°C, 20 cycles, n = 22 c = 0.
B4	1037	I_Z = column 8 of table III at T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750; for 2,000 cycles. Mounting conditions in accordance with 4.5.2.
B5	1027	I_Z = 40 percent of column 8 of table III minimum for 96 hours; Adjust T_A and / or I_Z to achieve T_J minimum. Temporary leads may be added for surface mount devices.
		Option 1: T_A = +100°C max; T_J = +275°C minimum; t = 96 hours. n =22, c = 0.
	or	Option 2: T_A = +30°C max; T_J = +175°C minimum; t = 1,000 hours n = 45, c = 0.
B6	3101 or 4081	$R_{\theta JL}$ = 42° C/W maximum; $R_{\theta JEC}$ = 20°C/W maximum; +25°C < T_A +35°C; reference temperature measuring point is the inside of mounting clip on lead or endcap (see 4.5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

* 4.4.2.2 Group B inspection, table VIB (JAN, JANTX and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
B2	1056	0°C to +100°C, 10 cycles, n = 22 c = 0.
B2	1051	-55°C to +175°C, 20 cycles, n = 22 c = 0.
B2	1071	Test condition E only NOTE: For non-transparent devices, hermetic seal may be performed after electrical measurements.
B3	1027	I_Z(min) = 50 percent of column 8 of table III minimum. Adjust I_Z to achieve T_J = 150°C min (see 4.5.7).

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E table VII of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, table I, subgroup 2 herein except $Z_{\theta JX}$ need not to be performed.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	0°C to +100°C, 10 cycles, n = 22 c = 0.
C2	1051	-55°C to +175°C, 20 cycles, n = 22 c = 0.
C2	2036	Tension - test condition A; 10 lbs; t = 15 s ±3 s. Lead fatigue - Test condition E. NOTE: Not applicable to US versions.
C2	1071	Test condition E only NOTE: For non-transparent devices, hermetic seal may be performed after electrical measurements.
C5	3101 or 4081	$R_{\theta JL}$ = 42° C/W maximum at L = .375 inch (9.53 mm) $R_{\theta JEC}$ = 20°C/W maximum for US types. n = 22, c = 0 (see 4.5.6).
C6	1027	I_Z(min) = 50 percent of column 8 of table III minimum. Adjust I_Z to achieve T_J = 150°C min (see 4.5.7).
C7	4071	Temperature coefficient for JAN, JANTX and JANTXV only; I_Z = column 5 of table III; T_{A1} = +25°C ±5°C; T_{A2} = +125°C ±5°C; limit = column 14 of table III (see paragraph 4.5.3).

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

4.5.1 Surge current (I_{ZSM}). The peak currents specified in column 10 of table III shall be applied in the reverse direction and shall be superimposed on the current (I_Z = column 5 of table III) a total of five surges at 1 minute intervals. Each individual surge shall be at one-half square wave pulse of 8.3 millisecond duration or an equivalent sine wave with the same effective (rms) current.

4.5.2 Voltage regulation (V_Z(reg)). A current of 10 percent of I_Z (table III, column 8) shall be maintained until thermal equilibrium is attained and the V_Z shall be noted. The current shall then be increased to a level of 50 percent of I_Z (table III, column 8) and maintained at this level until thermal equilibrium is attained at which time the voltage change shall not exceed table III, column 9. For this test, the diode shall be suspended by its leads (nonsurface mount) with mounting clips whose inside edge is located at 0.375 ±0.010 inch (9.52 ±0.25 mm) from the body and the lead temperature at inside edge of the mounting clips shall be maintained at a temperature between +23°C and +33°C. For surface mount packages, the diode shall be suspended by the endcaps with the temperature of the endcaps being maintained between +23°C and +33°C. For JANC, the die shall be stabilized at +25°C and the test shall be performed utilizing pulse condition. This measurement may be performed after a shorter time interval following application of the test current than that which provides thermal equilibrium if correlation can be established to the satisfaction of the qualifying activity.

4.5.3 Temperature coefficient of regulator voltage (α_{VZ}). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature.

4.5.4 Regulator voltage. The test current (column 5 of table III) shall be applied until thermal equilibrium is attained prior to reading the regulator voltage. For this test, the diode shall be suspended by its leads (non-surface mount) with mounting clips whose inside edge is located at .375 ± .010 inch (9.52 ±0.25 mm) from the body and the lead temperature at inside edge of the mounting clips shall be maintained at a temperature of +23°C to +33°C. For surface mount diodes, the diode shall be suspended by the end-caps with the temperature of the end-caps being maintained at +23°C to +33°C. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the qualifying activity. The breakdown voltage or JANHC and JANKC shall be read with a pulse measurement of 10 msec (max)

* 4.5.5 Pulse measurements. Conditions for pulse measurements shall be as specified in 4.3.2.2 of MIL-STD-750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

4.5.6 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3101 of MIL-STD-750. The following details shall apply:

a. I_H = 2.0 A dc minimum
b. I_M = 1 to 10 mA
c. t_{MD} = 100 μs maximum
d. t_H = thermal equilibrium

The device shall be allowed to reach thermal equilibrium at current I_H before the measurement shall be made.

Lead spacing: LS = .375 inches (9.53 mm) for leaded devices as defined on figure 3.
LS = 0 (end-cap mount) for US devices.

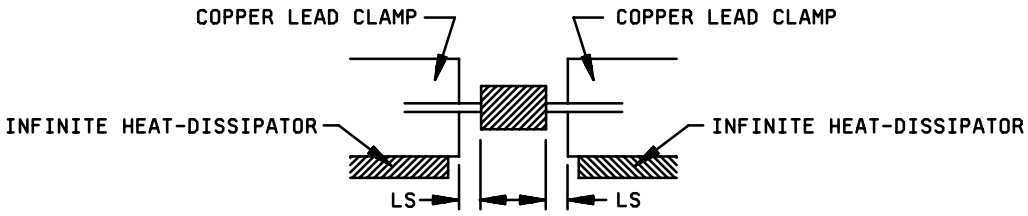


FIGURE 3. Mounting arrangement for thermal resistance measurements.

4.5.7 Free air burn-in. Deliberate heat sinking baffles to creat an over or forced air-cooling is prohibited unless otherwise approved by the qualifying activity. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period.

4.5.8 Scope display evaluation. Scope display evaluation shall be sharp and stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only, with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. The reverse current (I_{BR}) over the knee shall be 500 μA peak.

4.5.8.1 Scope display option. At the suppliers option, 100 percent scope display evaluation may be discontinued after three consecutive lots are 100 percent tested with zero failures. Any group A failure shall require 100 percent scope display to be reinvoked.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	Category I bond Category III bond (See 4.4.1)	$Z_{\theta JX}$		4.5 7.5	°C/W °C/W
Forward voltage	4011	I_F = 200 mA dc	V_{F1}		1.0	V dc
Forward voltage	4011	I_F = 1 A dc	V_{F2}		1.5	V dc
Reverse current leakage	4016	DC method; V_R = column 11 of table III	I_{R1}		Column 12	µA dc
Regulator voltage	4022	I_Z = column 5 of table III (see 4.5.5)	V_Z	Column 3 -5, -2, -1 percent	Column 4 +5, +2, +1 percent	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Reverse current leakage	4016	DC method; V_R = column 11 of table III	I_{R2}		Column 15	µA dc
* Subgroup 4						
Small-signal reverse breakdown impedance	4051	I_Z = column 5 of table III I_{sig} = 10 percent I_Z	Z_Z		Column 6	ohms
Knee impedance	4051	I_{ZK} = column 14 of table III I_{sig} = 10 percent I_{ZK}	Z_{ZK}		Column 7	ohms
Scope display evaluation	4023	See 4.5.8				

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection -Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_{ZSM} = column 10 of table III at T_A +25°C (see 4.5.1).	I_{ZSM}			
End-point electrical measurements		See table I, subgroup 2 except $Z_{\theta JX}$				
Subgroup 7						
Voltage regulation		See 4.5.2	$V_{Z(reg)}$		Column 9	V dc
Temperature coefficient of regulator voltage	4071	JANS level only I_Z = column 5 of table III T_{A1} = +25°C ±5°C, T_{A2} = 120°C ≤ T_2 ≤ 130°C	α_{VZ}		Column 13	%/°C

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

* TABLE II. Group E inspection (all quality levels).

Inspection 1/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Temperature cycling	1056	20 cycles, condition D except low temperature shall be achieved using liquid nitrogen (-195°C). Do a visual for cracked glass.	
Electrical measurements		See table I, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state intermittent operating life	1037	$I_Z = I_{Z2}$ (column 8 of table III) at T_A = room ambient for 10,000 cycles. No forced air cooling on the device shall be permitted. (Mounting conditions in accordance with 4.5.2.)	
Electrical measurements		See table I, subgroup 2	
Subgroup 3			3 devices, c = 0
DPA	2101	Cross section and scribe and break.	
Subgroup 4			
Thermal impedance curves		Each supplier shall submit their (typical) design thermal impedance curves. In addition, test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report.	
Subgroups 5 and 6			
Not applicable			
Subgroup 7			
Resistance to glass cracking	1057	Step stress to destruction by increased cycles or up to a maximum of 25 cycles.	n = 45
Subgroup 8			
Soldering heat	2031	One cycle	n = 45, c = 0

1/ A separate sample may be pulled for each task.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

TABLE III. Electrical characteristics and test conditions (all case outlines).

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15
Device type	V_Z Nom	V_Z Min 1/ 2/	V_Z Max 1/ 2/	I_Z test current T_A = +25°C	Z_Z Imped-ance	Z_K Knee imped-ance	I_Z Max dc current T_A = +25°C	V_Z(reg) voltage regula-tion 3/	I_{ZSM} T_A = +25°C 4/	V_R Revers voltage	I_R Revers current dc I_{R1}	α_{VZ} Tempe ature coeffi-cient 5/	I_{ZK} Test current	I_R Revers current T_A = +150°C I_{R2}
	V	V	V	mA	Ω	Ω	mA	V	A	V	µA	%/°C	mA	µA
1N6485	3.3	3.14	3.46	76	10	400	433	0.9	4.2	1.0	50.00	-.075	1.00	500
1N6486	3.6	3.42	3.78	69	10	400	397	0.8	3.9	1.0	50.00	-.070	1.00	200
1N6487	3.9	3.71	4.09	64	9	400	366	.75	3.6	1.0	35.00	-.060	1.00	100
1N6488	4.3	4.09	4.51	58	9	400	332	.70	3.3	1.0	5.00	-.050	1.00	100
1N6489	4.7	4.47	4.93	53	8	500	304	.60	3.0	1.0	4.00	±.025	1.00	100
1N6490	5.1	4.85	5.35	49	7	500	280	.5	2.7	1.0	1.00	±.030	1.00	100
1N6491	5.6	5.32	5.88	45	5	600	255	.40	2.5	2.0	0.50	±.040	1.00	100
1N4460	6.2	5.89	6.51	40	4	200	230	.35	2.3	3.72	10.00	+.050	1.00	50
1N4461	6.8	6.46	7.14	37	2.5	200	210	.30	2.1	4.08	5.00	+.057	1.00	20
1N4462	7.5	7.13	7.87	34	2.5	400	191	.35	1.9	4.50	1.00	+.061	0.50	10
1N4463	8.2	7.79	8.61	31	3.0	400	174	.40	1.7	4.92	0.50	+.065	0.50	5
1N4464	9.1	8.65	9.55	28	4.0	500	157	.45	1.6	5.46	0.30	+.068	0.50	3
1N4465	10	9.50	10.50	25	5.0	500	143	.50	1.4	8.0	0.30	+.071	0.25	3
1N4466	11	10.45	11.55	23	6.0	550	130	.55	1.3	8.8	0.30	+.073	0.25	2
1N4467	12	11.40	12.60	21	7.0	550	119	.60	1.2	9.6	0.20	+.076	0.25	2
1N4468	13	12.35	13.65	19	8.0	550	110	.65	1.1	10.4	0.05	+.079	0.25	2
1N4469	15	14.25	15.75	17	9.0	600	95	.75	.95	12.0	0.05	+.082	0.25	2
1N4470	16	15.20	16.80	15.5	10.0	600	90	.80	.90	12.8	0.05	+.083	0.25	2
1N4471	18	17.10	18.90	14	11.0	650	79	.83	.79	14.4	0.05	+.085	0.25	2
1N4472	20	19.00	21.00	12.5	12.0	650	71	.95	.71	16.0	0.05	+.086	0.25	2
1N4473	22	20.90	23.10	11.5	14	650	65	1.0	.65	17.6	0.05	+.087	0.25	2
1N4474	24	22.80	25.20	10.5	16	700	60	1.1	.60	19.2	0.05	+.088	0.25	2
1N4475	27	25.70	28.30	9.5	18	700	53	1.3	.53	21.6	0.05	+.090	0.25	2
1N4476	30	28.50	31.50	8.5	20	750	48	1.4	.48	24.0	0.05	+.091	0.25	2
1N4477	33	31.40	34.60	7.5	25	800	43	1.5	.43	26.4	0.05	+.092	0.25	2

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

TABLE III. Electrical characteristics and test conditions (all case outlines) - Continued.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15
Device type	V_Z Nom	V_Z Min 1/ 2/	V_Z Max 1/ 2/	I_Z test current T_A = +25°C	Z_Z Imped-ance	Z_K Knee imped-ance	I_Z Max dc current T_A = +25°C	V_Z(reg) voltage regula-tion 3/	I_{ZSM} T_A = +25°C 4/	V_R Revers voltag	I_R Rever curren dc I_{R1}	α_{VZ} Temper-ature coeffi-cient 5/	I_{ZK} Test current	I_R Reverse Current dc T_A = +150°C I_{R2}
	V	V	V	mA	Ω	Ω	mA	V	A	V	µA	%/°C	mA	µA
1N4478	36	34.2	37.8	7.0	27	850	40	1.7	.40	28.8	.05	+.093	0.25	2
1N4479	39	37.1	40.9	6.5	30	900	37	1.8	.37	31.2	.05	+.094	0.25	2
1N4480	43	40.9	45.1	6.0	40	950	33	1.9	.33	34.4	.05	+.095	0.25	2
1N4481	47	44.7	49.3	5.5	50	1000	30	2.1	.30	37.6	.05	+.095	0.25	2
1N4482	51	48.5	53.5	5.0	60	1100	28	2.3	.28	40.8	.05	+.096	0.25	2
1N4483	56	53.2	58.8	4.5	70	1300	26	2.5	.26	44.8	.25	+.096	0.25	10
1N4484	62	58.9	65.1	4.0	80	1500	23	2.7	.23	49.6	.25	+.097	0.25	10
1N4485	68	64.6	71.4	3.7	100	1700	21	3.0	.21	54.4	.25	+.097	0.25	10
1N4486	75	71.3	78.7	3.3	130	2000	19	3.3	.19	60.0	.25	+.098	0.25	10
1N4487	82	77.9	86.1	3.0	160	2500	17	3.6	.17	65.6	.25	+.098	0.25	10
1N4488	91	86.5	95.5	2.8	200	3000	16	4.0	.16	72.8	.25	+.099	0.25	10
1N4489	100	95.0	105.0	2.5	250	3100	14	4.4	.14	80.0	.25	+.100	0.25	10
1N4490	110	104.5	115.5	2.3	300	4000	13	5.0	.13	88.0	.25	+.100	0.25	10
1N4491	120	114.0	126.0	2.0	400	4500	12	5.5	.12	96.0	.25	+.100	0.25	10
1N4492	130	123.5	136.5	1.9	500	5000	11	6.0	.11	104	.25	+.100	0.25	10
1N4493	150	142.5	157.5	1.7	700	6000	9.5	7.0	.095	120	.25	+.100	0.25	10
1N4494	160	152	168	1.6	1000	6500	8.9	8.0	.089	128	.25	+.100	0.25	10
1N4495	180	171	189	1.4	1300	7000	7.9	10.0	.079	144	.25	+.100	0.25	10
1N4496	200	190	210	1.2	1500	8000	7.2	12.0	.072	160	.25	+.100	0.25	10

1/ See 4.5.5. Voltages shown are for 5 percent tolerance devices. Voltages for 2 and 1 percent tolerances devices shall be calculated accordingly.
2/ 1N4460D through 1N4496D and 1N6485D through 1N6491D are 1 percent voltage tolerance.
1N4460C through 1N4496C and 1N6485C through 1N6491C are 2 percent voltage tolerance.
1N4460 through 1N4496 and 1N6485 through 1N6491 are 5 percent voltage tolerance.
3/ See 4.5.2.
4/ See 4.5.1.
5/ See 4.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The lead finish as specified (see 3.6).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer’s List QML No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2689)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/406E	2. DOCUMENT DATE 8 September 2003

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICES, DIODE, SILICON, VOLTAGE REGULATOR TYPES 1N4460, THROUGH 1N4496, AND 1N6485, THROUGH 1N6491 1N4460US, THROUGH 1N4496US, AND 1N6485US, 1N6485CUS, 1N6485DUS THROUGH 1N6491US, PLUS C AND D TOLERANCE SUFFIXES; JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan.barone@dla.mil
c. ADDRESS Defense Supply Center, Columbus ATTN: DSCC-VAC, P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, Feb 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:37Z
Check the source to verify that this is the current version before use.